Help – Logs & Diagnostics

Menu: Help

Provides access to application logs, module diagnostics, and system information.

Log Files

Log	Description
Activity Log	Records application events and UDP/CAN communication activity in real time
Error Log	Records application errors and exceptions
Ethernet Log	Records raw UDP packet traffic — useful for diagnosing communication issues

Press **Start/Pause** to freeze or resume live log updates.

Activity Log — Module Status Events

When a module connects or its status changes, RC logs the following to the Activity Log automatically:

Log Entry	Description
Module X, Ethernet connected: True/False	Whether the module's Ethernet port is connected
Module X, Wifi Strength: 0–3	ESP32 WiFi signal level — 0 = not connected, 1 = weak (RSSI < −80), 2 = moderate (RSSI < −70), 3 = good (RSSI < −65)
Module X, Valves are 2 Wire / 3 Wire	Valve wiring mode reported by the module
Module X, Pin Configuration correct / not correct	Whether the module's pin configuration is valid

These entries are written once on first connect and again whenever the status changes.

Controls

Control	Action
Start/Pause log	Freeze or resume live log updates
Reset logs	Delete all log files after confirmation
PID Logging	Toggle PID diagnostic logging. While on (green), the module streams per-loop rate control data (target, applied rate, PWM, error) which is saved to a CSV file in the `PIDLogs` folder inside the data folder — useful for tuning and diagnosing rate control problems. The setting is remembered between sessions.
Open rate graph	Display the rate graph
Check for updates	Check for a newer version of the application
Open user manual	Open the user manual
Navigate modules left	View previous module diagnostics
Navigate modules right	View next module diagnostics

Module Diagnostics

Display	Description
Module ID	ID of the module currently being viewed (use left/right arrows to switch)
Firmware Version	Firmware version running on the selected module
Elapsed Time	Seconds since last module data was received

System Information

Display	Description
IP / Subnet	Current Ethernet subnet in use
Application Version	Installed RC software version and date
Current File	Name of the active profile file
GPS Coordinates	Live latitude and longitude from AgOpenGPS

External Links

Links to the project repositories are available from this screen:

- AOG RC GitHub repository
- PCB Setup GitHub
- Rate Control PCBs GitHub

Troubleshooting

No Relays Activating

1. Is the module Config set correctly for board type and relay type?
2. Has the subnet been set in the app and uploaded to the module?
3. Have section count and widths been set?
4. Is Auto mode on?
5. Is a section assigned to each relay?
6. Is a switch assigned to each section?
7. If Auto is on, have sensor counts/unit and base rate been set?
8. If Work Switch is enabled (Menu > Switches), the master will only activate when the work switch reads on — check the switch's wiring, pin assignment, and the **Invert Work Switch** setting (Modules > Pins) against whether the sensor is normally-open or normally-closed.

No UPM (No Flow Reading)

1. Check ground speed.
2. Check working width.
3. Check sensor control settings.
4. Is AgOpenGPS connected?
5. Is the module connected?
6. Check sensor counts/unit.
7. Check base rate.
8. Have control values been sent to the module?
9. Check control type — is it a valve or motor?
10. Is the connector fully plugged in?
11. Is the module running the latest firmware?
12. Is Auto mode on?
13. If Work Switch is enabled, is it reading on? A wrong **Invert Work Switch** setting (Modules > Pins) will hold the master off with no flow, even though everything else is configured correctly.

Document Applied Manual Rate (Mode 5)

1. Don't assign a real module to a product set to Mode 5 — the module's real sensor data will conflict with the manually entered value.
2. Enter the flow value in units per minute, not per acre/hectare; there is no rate alarm in this mode to catch a units mistake.
3. Applied rate will rise and fall with ground speed even though the flow input is constant — this is expected for a fixed-metering implement.
4. A ground speed source is still required — GPS (via AgOpenGPS), a wheel-speed sensor module, or a manually entered/simulated speed (Menu > Speed) all work; with no speed source active, quantity won't accrue even though the flow value is set.

No Rate Adjustment

1. In manual mode, the master switch must be on.
2. Is minimum PWM too low?

Motor Won't Shut Off

1. Minimum PWM should be set to 0.
2. Is master override enabled?

After Updating Module Firmware

1. Resend the subnet (Modules > Communication).
2. Resend all module config settings (Modules > Config, then Send).